Per-zero exclusion bounds free of assumptions on the zeros above the band, sharpened by a formally verified zero-free half-plane

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8 October 2026 — version 0.4.1

Abstract. A positive value λ of the Weil quadratic form, truncated to the window $[-h, h]$, $h = \frac{1}{2}\log 13$, evaluated at its pole-conditioned minimiser and computed on the geometric side independently of the zeros, is converted into a bound $|\operatorname{Re}\rho_k - \frac{1}{2}| \le \delta_k^*$ for individual zeros ρ_k of $\zeta(s)$, with no assumption on the location of the infinitely many zeros above the band; the other in-band zeros enter at their on-line ordinates, a range verified rigorously by Platt and Trudgian. The price of this independence is an envelope of the explicit formula taken over the whole width of the region in which zeros may lie. We feed into this envelope the zero-free half-plane $\operatorname{Re} s > 7/8$ of OpenAI (2026), whose Lean formalisation we compiled independently from a fresh clone and audited (7 061 modules; axioms `propext`, `Classical.choice`, `Quot.sound` only; negative control detected). The admissible width drops from $\frac{1}{2}$ to $\frac{3}{8}$, the envelope factor from 13.20 to 6.04, and every bound tightens by the same factor 1.501. In the cell $q = 13$, $\Omega = 75.55$, $d = 21$ we obtain $|\operatorname{Re}\rho_1 - \frac{1}{2}| \le 3.50\times10^{-19}$, $|\operatorname{Re}\rho_5 - \frac{1}{2}| \le 9.93\times10^{-15}$, $|\operatorname{Re}\rho_{10} - \frac{1}{2}| \le 4.31\times10^{-10}$, $|\operatorname{Re}\rho_{18} - \frac{1}{2}| \le 5.74\times10^{-3}$. The envelope is built on a two-dimensional grid across the whole admissible strip, its Abel summation includes the boundary term, and condition (C2) is computed at 120 digits; the one remaining input, the contribution (A) of the zeros above $G_2 = 2000$, is quantified by doubling G_2 at equal grid spacing: it changes the envelope by 6.4×10^{-4} and the first bound by 3.4×10^{-4} relative. The cost of independence from the zeros above the band, measured as the ratio to the conditional bound, falls from 4.36 to 2.90. To our knowledge this is the first use of a formally verified zero-free region as an input to a finite positivity certificate.

Keywords: Riemann zeta function; Weil explicit formula; zero-free region; per-zero exclusion; positivity certificate; Lean formal verification.

1 Introduction and contribution

In earlier notes [5, 6, 7] a chain was built from one finite object to statements about individual zeros: a truncated Weil form, its positive minimum λ under the pole condition, the silent subspace in which the minimiser annihilates the in-band zeros, and from these a bound on the distance of a chosen zero from the critical line. Version 0.3 [7] removed the hypothesis that only one zero leaves the line, by a single inequality (Lemma 1 below), for all zeros above the band. The entire cost of that step sits in one quantity: an upper envelope U_{top} of the explicit-formula contributions of all those zeros, wherever they may lie. That envelope must be taken across the full admissible width κ of the zero region, and its size grows rapidly with κ.

The present note makes three contributions.

1. *A formally verified global input enters a local certificate.* The zero-free half-plane $\operatorname{Re} s > 7/8$ [1, 2] reduces the admissible width from $\kappa = \frac{1}{2}$ to $\kappa = \frac{3}{8}$ (Proposition 2). The machinery is unchanged; only the height of the envelope is moved. Every bound improves by the factor $\sqrt{B_{1/2}/B_{3/8}} = 1.501$ (Table 4).

2. *Independent formal verification* of that input on the author's own hardware from a fresh clone, with axiom audit, negative control and statement check (Section 5, Table 1).

3. *Every input stated and measured.* The vertical behaviour of $|F(x+iy)|^2$ is governed by the expansion (3.4) (Remark 3); the envelope is therefore built on a two-dimensional grid, and the property (E) that its upper edge dominates the strip is verified with no exceedance. The Abel summation carries its boundary term (3.3). The scope of the envelope (zeros above the band) and the role of the in-band zeros (condition (I)) are stated separately, and every input of the final bound is classified in a ledger (Table 5). Version 0.4 stated (E) and (I); version 0.4.1 measures (E), adds the boundary term and measures (A) at equal grid spacing.

2 Setting and the basic inequality

Notation follows [5, 7]. For a real even η supported in $[-h,h]$ put $F(z)=\int \eta(u)\cos(zu)\,du$; F is entire and real on the real axis, so $F(\bar z)=\overline{F(z)}$. For the minimiser η_0 the explicit formula expresses the truncated form as a sum over all nontrivial zeros $\rho_j=\frac12+\delta_j+i\gamma_j$ ($\gamma_j>0$, with multiplicity),

$$\lambda=\sum_j t_j,\qquad t_j=2\,\mathrm{Re}\,F(\gamma_j+i\delta_j)^2, \tag{2.1}$$

so that $t_j=2F(\gamma_j)^2\ge 0$ for a zero on the line. The value λ is computed on the geometric side (primes, pole, archimedean term) and uses no zero.

Lemma 1. *If $U_{\mathrm{top}}\ge\sum_j\max(t_j,0)$, then for every zero k*

$$-t_k\le U_{\mathrm{top}}-\lambda=:B. \tag{2.2}$$

Proof. $\lambda=\sum_j t_j\le\sum_{j\ne k}\max(t_j,0)+t_k\le U_{\mathrm{top}}-\max(t_k,0)+t_k\le U_{\mathrm{top}}+t_k$. □

The lemma itself places no condition on the zeros $j\ne k$; everything rests on U_{top}. Split the zeros into the $N(\Omega)=18$ zeros with $\gamma_j<\Omega$ (the band) and those with $\gamma_j>\Omega$. For the band we use

(I) the in-band zeros with ordinate different from γ_k lie on the line,

a statement contained in the rigorous verification of Platt and Trudgian up to height $3\cdot10^{12}$ [8]. Their positive parts then sum to at most the computed in-band total $S_{\mathrm{band}}=1.28\times10^{-49}$. For the zeros above the band no assumption is made: their contributions are bounded by the envelope of the next section.

3 The width of the envelope

Proposition 2. *Every nontrivial zero ρ of ζ satisfies*

$$|\,\mathrm{Re}\,\rho-\tfrac12|\le\tfrac38. \tag{3.1}$$

Proof. By [1], formalised in Lean as `OAI.riemannZeta_ne_zero_of_seven_eighths_lt_re` [2] and re-verified in Section 5, $\zeta(s)\ne0$ for $\mathrm{Re}\,s>7/8$. The functional equation maps a zero ρ to the zero $1-\bar\rho$; applying the half-plane to both gives $\mathrm{Re}\,\rho\le7/8$ and $1-\mathrm{Re}\,\rho\le7/8$. □

For an off-line zero, $\max(t_j,0)\le2|F(\gamma_j+i\delta_j)|^2\le2\widehat\Phi_\kappa(\gamma_j)$ with

$$\widehat\Phi_\kappa(g)=\max_{0\le\delta\le\kappa}|F(g+i\delta)|^2,\qquad\kappa=\tfrac38, \tag{3.2}$$

using $|F(g-i\delta)|=|F(g+i\delta)|$. The bound U_5 of [6] (decreasing running-maximum envelope, Abel summation, the unconditional zero-counting estimate of Hasanalizade–Shen–Wong [4] entering once) applies verbatim to any non-negative function of the ordinate, because the

counting estimate does not depend on real parts. Built on (3.2) over $\Omega < g \le G_2$ it bounds $\sum_{\Omega<\gamma_j\le G_2} \max(t_j, 0)$; together with the tail (A) of the zeros above G_2, $U_{\mathrm{top}} = U_5^{(\kappa)} + 2S_{\mathrm{band}}$ is admissible in Lemma 1. Explicitly, with the grid $\Omega = g_0 < \cdots < g_n = G_2$, the running maximum $M(g_i) = \max_{j\ge i} \widehat{\Phi}_\kappa(g_j)$, and $T(g) = N_{\mathrm{d}}(g) - N_{\mathrm{d}}(\Omega) + 2R(g)$, where N_{d} is the smooth zero-counting function and R the bound of [4],

$$U_5^{(\kappa)} = 2\Big[\sum_{i<n} \big(M(g_i) - M(g_{i+1})\big)\, T(g_{i+1}) \;+\; M(G_2)\, T(G_2)\Big]. \tag{3.3}$$

The last term counts the floor $M(G_2)$ for the zeros in $(\Omega, G_2]$; it contributes 7.70×10^{-4} of U_5 for the unshifted envelope and 3.66×10^{-4} for $\kappa = \frac{3}{8}$. The tail (A) is measured in Section 6 (Table 2).

Remark 3. The upper edge $\Phi_\kappa(g) = |F(g + i\kappa)|^2$ is the natural candidate for the envelope; its link to the segment envelope (3.2) is the property

(E) $\widehat{\Phi}_\kappa(\gamma_j) \le \sup_{t\ge\gamma_j} \Phi_\kappa(t)$ for every ordinate γ_j above the band.

Near the real axis

$$|F(x + iy)|^2 = F(x)^2 + y^2\big(F'(x)^2 - F(x)F''(x)\big) + O(y^4), \tag{3.4}$$

so the vertical profile rises where $F'^2 > FF''$ and dips where $FF'' > F'^2$, for instance at a local minimum of $|F|$ with $F \ne 0$. For random even Legendre combinations of degree ≤ 14 on $[-h, h]$ the segment maximum exceeds the running-maximum edge envelope at grid points by up to 12%. The computation of this note therefore builds the envelope directly on the segment maximum (3.2), evaluated on a grid of 8001 points in g times 9 points $\delta \in \{0, \kappa/8, \ldots, \kappa\}$, and reports (E) as a by-product: for η_0 the segment envelope coincides with the edge envelope at every grid point (maximal ratio 1.0, no exceedance), for $\kappa = \frac{1}{2}$ and for $\kappa = \frac{3}{8}$.

4 The per-zero bound

For a silent ordinate γ_k (in-band, $F(\gamma_k)^2$ below the in-band total 1.28×10^{-49}) define

$$g_k(\delta) = 2F(\gamma_k)^2 - 2\,\mathrm{Re}\,F(\gamma_k + i\delta)^2, \qquad C_k = \lim_{\delta\to 0} g_k(\delta)/\delta^2. \tag{4.1}$$

Theorem 4. *Assume (I). Let $\lambda > 0$, let $U_{\mathrm{top}} \ge 2S_{\mathrm{band}} + \sum_{\gamma_j>\Omega} 2\widehat{\Phi}_\kappa(\gamma_j)$ with $\kappa = \frac{3}{8}$, and $B = U_{\mathrm{top}} - \lambda$. If g_k is positive and non-decreasing on $(0, \kappa]$ (C2), then every zero with ordinate γ_k satisfies $|\,\mathrm{Re}\,\rho - \frac{1}{2}| \le \delta_k^*$, where*

$$g_k(\delta_k^*) = B + 2F(\gamma_k)^2, \qquad \delta_k^* = \sqrt{B/C_k}\,\big(1 + O(\delta_k^{*2}) + O(F(\gamma_k)^2/B)\big), \tag{4.2}$$

and $\delta_k^* = \kappa$ if this equation has no root in $(0, \kappa]$.

Proof. Let $\rho = \frac{1}{2} + \delta + i\gamma_k$ be a zero. By Proposition 2 every zero above the band has $|\delta_j| \le \kappa$, so its positive part is at most $2\widehat{\Phi}_\kappa(\gamma_j)$; by (I) the in-band zeros at the other ordinates contribute at most S_{band}. For the tested zero and, if it is off the line, its reflection $1 - \bar\rho$ (same ordinate, same contribution), $t_k = 2F(\gamma_k)^2 - g_k(|\delta|) \le 2F(\gamma_k)^2$ by (C2); together with the other in-band zeros the band contributes at most $S_{\mathrm{band}} + 2F(\gamma_k)^2 \le 2S_{\mathrm{band}}$. Hence $U_{\mathrm{top}} \ge \sum_j \max(t_j, 0)$ and Lemma 1 gives $-t_k \le B$. Since $-t_k = g_k(|\delta|) - 2F(\gamma_k)^2$, we get $g_k(|\delta|) \le B + 2F(\gamma_k)^2$, and monotonicity on $(0, \kappa]$ gives $|\delta| \le \delta_k^*$. □

In the cell below $S_{\mathrm{band}}/B \le 5.0 \times 10^{-10}$; the terms $2S_{\mathrm{band}}$ in U_{top} and $2F(\gamma_k)^2$ in (4.2) change δ_k^* by at most 10^{-9} relative. The tabulated values are $\sqrt{B/C_k}$ with $B = U_5^{(\kappa)} - \lambda$.

5 Independent formal verification of the input

The Lean development of [2] (commit `adc7f1241`) was compiled from a fresh clone under WSL Ubuntu on the author's workstation, with the toolchain and dependency revisions pinned by the repository. The repository's own sequence `lake update` → `lake exe cache get` → `lake build` is required: `lake update` applies twelve post-resolution Lean-4.34.1 compatibility patches from `patches/` (among them `rellich-kondrachov`, `PrimeNumberTheoremAnd`, `StrongPNT`); without them three modules of `rellich-kondrachov` fail to elaborate. Native Windows checkouts additionally meet path-length, line-ending and file-name (`:`) limits. An independent Comparator re-run is reported in [3].

Table 1: Independent build and audit of the formal input (8 October 2026).

Toolchain	Lean 4.34.1; Mathlib `d13f23b7`; 8 908 cache files
Build target	`OAI.NumberTheory.DirichletL.Nonvanishing`
Result	`Build completed successfully (7061 jobs)`
Axioms, ζ theorem	`propext`, `Classical.choice`, `Quot.sound`
Axioms, Dirichlet theorem	`propext`, `Classical.choice`, `Quot.sound`
Negative control	`2+2=5 := by sorry` ⇒ `[sorryAx]` (detected)
Statement check	with `pp.fullNames`: $\forall s,\ 7/8 < \operatorname{Re} s \to$ `riemannZeta` $s \neq 0$, Mathlib's own `riemannZeta`

6 Measurements

Cell and tool. $q = 13$, $h = 1.2824746787$, $\Omega = 75.5524$, $N = 18$ in-band zeros, $d = 21$, 300 ordinates, $G_2 = 2000$, envelope grid $n = 8000$ in g and 9 values of δ. $\lambda = \lambda_0^c = 1.44769516664 \times 10^{-41}$ at 120 digits, reproduced at 100 digits to 1.2×10^{-61} relative (precision floor 2.2×10^{-118}). Engine `sinir6.py` (md5 `d121415ee20ad874026faab9542e4d29`), patched by `yama_kayma_v3.py` (md5 `56ec393c56511b311865b38470ff3d9e`) into `sinir6_kayma_v3.py` (md5 `527c3a8d05a5f5ad7f1007945e023739`, stamped by the engine in each log), with options for the width κ, the segment grid and the boundary term; logs `kayma_v3_yarim_tam.log`, `kayma_v3_38_tam.log`. Gates in both runs: KAPI-0 15/15, KAPI-SINIR 4/4, KAPI-U5 3/3. Regression: before the boundary term is added, the patched engine reproduces version 0.4 to all printed digits ($5.9374883 \times 10^{-40}$, $2.7161074 \times 10^{-40}$, $4.4963024 \times 10^{-41}$).

Table 2: Envelope and budget, Abel boundary term included. $U_5 = 4.4997651 \times 10^{-41}$ is the unshifted bound ($U_5/\lambda = 3.108$).

	conditional (U_5)	$\kappa = \frac{1}{2}$	$\kappa = \frac{3}{8}$ (this note)
U_{top}	$4.4997651 \times 10^{-41}$	$5.9390539 \times 10^{-40}$	$2.7171004 \times 10^{-40}$
U_{top}/U_5	1	13.1986	6.03832
$\cosh^2(h\kappa)$	—	1.47073	1.24968
$B = U_{\mathrm{top}} - \lambda$	$3.0520699 \times 10^{-41}$	$5.7942844 \times 10^{-40}$	$2.5723309 \times 10^{-40}$
boundary term share	7.70×10^{-4}	2.64×10^{-4}	3.66×10^{-4}
(E) exceedances on the grid	—	0	0
(A) share, $G_2 \to 2G_2$	1.18×10^{-3}	—	6.40×10^{-4}

(C2) for η_0. Computed on a grid in $\delta \in (0, \frac{1}{2}] \supset (0, \frac{3}{8}]$ at 120 digits (`sinir7.py`, md5 `1cf751b20ab483ca760f35106a469294`), with $r(\delta) = g_k(\delta)/(C_k\delta^2)$; the grid method detects the sign turn of the synthetic negative control $F(z) = (z^2 - a^2)e^{-z^2/2}$, $a = 0.2$ [7].

Table 3: Condition (C2) for the minimiser η_0.

k	$r(0.1)$	$r(0.25)$	$r(0.5)$	$\min r$	sign turn / monotonicity break
1	0.997369	0.983552	0.934118	0.934118	none / none
5	1.000500	1.002950	1.009050	1.000010	none / none
10	1.003330	1.020540	1.077790	1.000030	none / none
18	0.987146	0.918537	0.658386	0.658386	none / none

Table 4: Per-zero bounds $\delta_k^* = \sqrt{B/C_k}$. Conditional: one off-line zero at a time. Columns κ: no assumption on the zeros above the band, (I) in the band.

k	γ_k	C_k	conditional	$\kappa = \frac{1}{2}$	$\kappa = \frac{3}{8}$
1	14.134725	2.09871×10^{-3}	1.20593×10^{-19}	5.25441×10^{-19}	$\mathbf{3.50096 \times 10^{-19}}$
5	32.935062	2.60878×10^{-12}	3.42041×10^{-15}	1.49032×10^{-14}	$\mathbf{9.92989 \times 10^{-15}}$
10	49.773832	1.38716×10^{-21}	1.48332×10^{-10}	6.46304×10^{-10}	$\mathbf{4.30626 \times 10^{-10}}$
18	72.067158	7.79709×10^{-36}	1.97848×10^{-3}	8.62052×10^{-3}	$\mathbf{5.74377 \times 10^{-3}}$
ratio to conditional (cost of independence)				4.357	**2.903**
gain from the formal half-plane, $\sqrt{B_{1/2}/B_{3/8}}$					**1.501**

The four ratios $\delta_{1/2}^*/\delta_{3/8}^*$ agree to five digits (1.5008), as they must, since only B changes. Each δ_k^* was recomputed from B and C_k independently of the engine; the largest deviation from the printed value is 3.8×10^{-6} relative, the rounding of C_k. All bounds satisfy $\delta_k^* < \frac{3}{8}$.

7 Ledger of the inputs

Table 5: Status of every input of the bound $|\operatorname{Re}\rho_1 - \frac{1}{2}| \le 3.50 \times 10^{-19}$.

Lemma 1, Theorem 4	proved here
Zero-free half-plane $\operatorname{Re} s > 7/8$	formally verified (Lean), re-verified independently
Width $\kappa = \frac{3}{8}$ (Proposition 2)	above + functional equation
Zero counting $R(T)$ in U_5	theorem [4]
λ	computed, two precisions, 1.2×10^{-61} agreement
Segment envelope, grid 8001×9	computed; Abel boundary term included (3.3)
(C2) for η_0	computed on a grid, 120 digits
(A) zeros above $G_2 = 2000$	computed: doubling G_2 at equal grid spacing changes U_{top} by 6.4×10^{-4} ($\kappa = \frac{3}{8}$); including it moves δ_1^* to 3.5021×10^{-19}
(E) vertical domination	computed: no exceedance on the grid, $\kappa = \frac{1}{2}$ and $\frac{3}{8}$
(I) in-band zeros at the other ordinates on the line	theorem, rigorous verification to $3 \cdot 10^{12}$ [8]

8 Author's contribution

The construction of the per-zero chain (silent subspace, pole-conditioned minimiser, overflow regime, envelope bound U_5), the removal of the single-violation hypothesis through Lemma 1, and the use of a global zero-free region as the height of the local envelope are the author's. The reproduction of the formal input, the engine runs and their regression checks were carried out

on the author's workstation. The ordinates used lie in the range verified rigorously by Platt and Trudgian [8], which serves as an external consistency check; the contribution is the mechanism by which one finite certificate yields a separate bound for each zero while the infinitely many zeros above the band carry no assumption.

9 Continuation

Next: the overflow cell $\Omega = 103.73$, $d = 29$, where the conditional bound is 1.56×10^{-23} and the cell lies inside the Schoenfeld range used by the prime-counting certificate; an analytic decay bound for the envelope beyond G_2, which turns (A) into a theorem; and the argument-principle count gate for ordinates between the targets.

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